Chapter 8

MANAGING DISTRIBUTED PARALLEL WORKFLOW SYSTEMS USING A MULTI-AGENT METHOD

Samir Aknine
Lip6, OASIS, Université Paris 6
8, rue du Capitaine Scott
75015 PARIS Cedex 15
France
Samir.Aknine@lip6.fr

Suzanne Pinson
Lamsade, Université Paris 9
Place Du Maréchal De Lattre
75775 PARIS Cedex 16
France
Suzanne.Pinson@lamsade.dauphine.fr

Abstract: In this chapter, we introduce new research directions for agent knowledge representation and multi-agent research and we enhance distributed parallel workflow systems (DPWS) with theory and tools developed in Artificial Intelligence. This article shows the limits of the traditional workflow systems and proposes a way of overcoming them through a new workflow model which parallelizes its activities. The first section defines the concepts of task, task execution validity, dependence and consistency between tasks. The second presents a multi-agent model which is used to manage the interactions between human agents, called actors, and artificial agents. It goes on to present a multi-agent software agent architecture which addresses the needs of these systems. This architecture is composed of several primitive agents and is innovative in the context of workflow systems. It offers several advantages: parallelization of agent tasks, reuse of agent components and partial mobility of agent code. Finally, the model is illustrated on a real application, that of technical specification cooperative writing in the domain of telecommunications.

Key words: agents, distributed workflow

1. INTRODUCTION

Research on designing cooperative work aims to build two kinds of systems: those able to assist a user in the resolution of a problem and those

Y. Ye et al. (eds.), *Agent Supported Cooperative Work*
© Kluwer Academic Publishers 2003

which coordinate the activity of a collective workgroup in an intelligent way. Work on systems which mediate cooperative activities comes under CSCW (Computer Supported Cooperative Work) in which several approaches exploit the concepts of Distributed Artificial Intelligence and contribute in return to enrich this discipline (Aknine, 1999; Singh, 1998). Some of this work, research on workflow (Borghoff et al., 1997; Gupta et al., 1996) addresses these problems well. The well-known definition of a workflow is as follows. A workflow is a formal representation of organizational activity processes performed by actors, i.e. human being, who participate in cooperative work. These actors may work on several decentralized sites (Schal, 1996). We know that workflow systems are the first applications to have offered mechanisms for the execution of activity processes and for checking these processes at the same time. De Michelis and Grasso define a workflow application as a tool for the specification and automation of cooperative activity processes (De Michelis and Grasso, 1993). Workflow systems do not only specify activity plans but they also help to execute, check and coordinate dynamically groups of actors which participate in the execution of the activities (Schal, 1996). However, the use of Internet by actors today has introduced new difficulties into the management of cooperative work. These difficulties are inherent to the distribution of cooperative applications. This makes the task of coordinating the activities of these different actors increasingly difficult. Considering the way in which workflow systems are defined, we have sought to give them all the flexibility of multi-agent methods. Our approach distributes the control of these systems among several software agents which ensure that the activities of the workflows will be well coordinated, executed and verified.

This article is part of a research project which aims to build a multi-agent approach for designing intelligent workflow systems. These new systems can adjust their behavior dynamically while facilitating the parallelization and scheduling of certain actor tasks. These systems offer coordination mechanisms which can ensure an active aid for the actors during the execution of their tasks and can propose an efficient assignment of their tasks. This article presents our work on the design of agent-oriented parallel workflow systems. In particular, the following aspects are addressed: (1) we propose a task representation model for flexible workflow process activities; (2) we develop our method for a distributed management of cooperative activities using artificial agents; (3) we propose an agent architecture which is based on the functionalities of the cooperative work processes; (4) we develop the cooperative application we have considered in order to illustrate and test our method. Compared to other workflow approaches, we have not only focused our research on the development of a workflow system but we have also defined a new agent architecture and a knowledge representation model for designing these systems. Generally, workflow research focuses on modeling the activity processes (Gupta et al., 1996). However, these